

Lab 12: Email forensics

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Image “cfreds_2015_data_leakage_pc.dd” from Lab 5

Goals

- Email forensics (Application Layer)

Step 1

What application was used for e-mail communication?

```
giavan@giavan-VirtualBox:~$ rip.pl -r SOFTWARE -p install | grep -i outlook
Launching installer v.20200517
20150322 - Microsoft Outlook MUI (English) 2013 15.0.4420.1017 (Microsoft Corporation)
giavan@giavan-VirtualBox:~$
```

List of email clients

- https://en.wikipedia.org/wiki/Comparison_of_email_clients

What was the e-mail account used by the suspect?

```
fls -rF -o 206848 cfreds_2015_data_leakage_pc.dd | grep -P "\.ost\$"
```

```
fls -rF -o 206848 cfreds_2015_data_leakage_pc.dd | grep -E "\.ost$"
```

```
giavan@giavan-VirtualBox:~/Downloads$ cd ~/Downloads
fls -rF -o 206848 cfreds_2015_data_leakage_pc.dd | grep -E "\.ost$"
r/r 46112-128-4:      Users/informant/AppData/Local/Microsoft/Outlook/iaman.informant
@nist.gov.ost
giavan@giavan-VirtualBox:~/Downloads$
```

Step 2.

List all e-mails of the suspect. If possible, identify deleted e-mails

- Install email extracting tool: *libpff*
- Copy *.ost* file from a DD image
- Extract email via *libpff*
- Install *pffexport* tool

```
giavan@giavan-VirtualBox:~/Downloads$ sudo apt update
sudo apt install pff-tools
Hit:1 http://vn.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://vn.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://vn.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

- Verify installation

```
giavan@giavan-VirtualBox:~/Downloads$ pffexport -h
pffexport 20180714

Use pffexport to export items stored in a Personal Folder File (OST, PAB
and PST).

Usage: pffexport [ -c codepage ] [ -f format ] [ -l logfile ] [ -m mode ]
               [ -t target ] [ -dhqvV ] source
```

- Copy email to current directory

```
giavan@giavan-VirtualBox:~$ cp /media/root/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Outlook/iaman.informant\@nist.gov.ost .
giavan@giavan-VirtualBox:~$ ls -l iaman.informant\@nist.gov.ost
-rwxr-xr-x 1 root root 16818176 Nov 25 15:10 iaman.informant@nist.gov.ost
```

- Extract emails via *pffexport*

```
giavan@giavan-VirtualBox:~$ pffexport iaman.informant\@nist.gov.ost
pffexport 20180714

Opening file.
```

- Verify the default output directory

```
giavan@giavan-VirtualBox:~$ ls -l iaman.informant\@nist.gov.ost.export/
total 8
drwxr-xr-x 12 root root 4096 Nov 25 15:25 'Root - Mailbox'
drwxr-xr-x  4 root root 4096 Nov 25 15:25 'Root - Public'
```

- Access a message in suspect's Mailbox

```
ls -l iaman.informant\@nist.gov.ost.export/Root\ -\ Mailbox/
```

```
giavan@giavan-VirtualBox:~$ ls -l iaman.informant\@nist.gov.ost.export/Root\ -\ Mailbox/
total 40
drwxr-xr-x  2 root root 4096 Nov 25 15:25 'Common Views'
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Drizzle
drwxr-xr-x  8 root root 4096 Nov 25 15:25 Finder
drwxr-xr-x 19 root root 4096 Nov 25 15:25 IPM_SUBTREE
```

- Access a message in suspect's *IPM_SUBTREE*

```
giavan@giavan-VirtualBox:~$ ls -l iaman.informant\@nist.gov.ost.export/Root\ -\ Mailbox/IPM_SUBTREE/
total 68
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Calendar
drwxr-xr-x  8 root root 4096 Nov 25 15:25 Contacts
drwxr-xr-x  2 root root 4096 Nov 25 15:25 'Conversation Action Settings'
drwxr-xr-x  6 root root 4096 Nov 25 15:25 'Deleted Items'
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Drafts
drwxr-xr-x  7 root root 4096 Nov 25 15:25 Inbox
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Journal
drwxr-xr-x  2 root root 4096 Nov 25 15:25 'Junk Email'
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Notes
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Outbox
drwxr-xr-x  2 root root 4096 Nov 25 15:25 'Quick Step Settings'
drwxr-xr-x  2 root root 4096 Nov 25 15:25 'RSS Feeds'
drwxr-xr-x  4 root root 4096 Nov 25 15:25 'Sent Items'
drwxr-xr-x  2 root root 4096 Nov 25 15:25 SmsAndChatsSync
drwxr-xr-x  8 root root 4096 Nov 25 15:25 'Sync Issues'
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Tasks
giavan@giavan-VirtualBox:~$ 096 Nov 25 15:25 'Working Set'
```

```
ls -l iaman.informant\@nist.gov.ost.export/Root\ -\ Mailbox/IPM_SUBTREE/Inbox/
```

```
giavan@giavan-VirtualBox:~$ ls -l iaman.informant\@nist.gov.ost.export/Root\ -\ Mailbox/IPM_SUBTREE/Inbox/
total 20
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Message00001
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Message00002
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Message00003
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Message00004
drwxr-xr-x  2 root root 4096 Nov 25 15:25 Message00005
```

```
cat iaman.informant\@nist.gov.ost.export/Root\ -\
Mailbox/IPM_SUBTREE/Inbox/Message00001/OutlookHeaders.txt | grep -Ei
"time|subject|name"
grep -E 'strings' Message*
```

```
giavan@giavan-VirtualBox:~$ ls -l iaman.informant\@nist.gov.ost.export/Root\ -\ Mailbox
/IPM_SUBTREE/Inbox/Message00001/OutlookHeaders.txt | grep -Ei "time|subject|name"
Client submit time:      Mar 23, 2015 17:29:27.000000000 UTC
Delivery time:          Mar 23, 2015 17:29:29.181071700 UTC
Creation time:          Mar 23, 2015 17:29:29.087320500 UTC
Sender name:            spy
Sent representing name: spy
```

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 12.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 12 From YOUR NAME**", replacing "YOUR NAME" with your real name.